Lecture 14: Labor, Incentives, and Data Externalities

CS120: Introduction to AI Safety - November 10, 2026, Zachary Robertson

Selected questions: Will, Paulina, Hudson, Yujen, Jolie, Orhun, and Justin

From action choices to data-production loops

Module 2	Module 3	Module 4	Module 5
Model behavior	**Agent policy**	**Interaction process**	**Institutional ecosystem**
shortcut learning	reward hacking	sycophancy	benchmark gaming
miscalibration	specification gaming	feedback-loop amplification	endogenous evaluation

Course sequence, not a hierarchy

Interactions with users and **workers** also change future data.

Who produces data, and who gets recognized?

Commissioned: experts, contractors, crowdworkers.

Not commissioned: end-user feedback and public knowledge contributions.

Producing useful data is not the same as being recognized or compensated as a worker.



Roadmap

What can this data establish?

Who bears the cost of producing it?

How does AI change future data?

Who made the reference changes what it tells us

Consider a reference used to evaluate medical answers:

A domain expert provides a diagnosis.

A user gives a chat response a thumbs-up.

A crowdworker submits an AI-assisted answer.

Who produced it, and what judgement were they asked to make?

Estimating LLM assistance in crowdwork

June 2023 MTurk study: summarize medical abstracts in about 100 words.

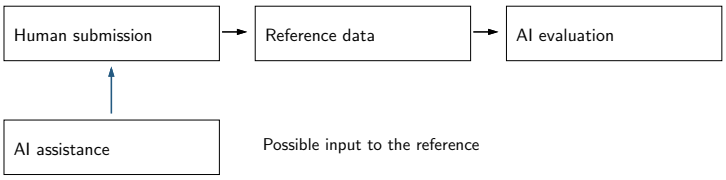
46 submitted summaries from 44 workers.

33-46%

Estimated share of summaries incorporating LLM assistance.

Veselovsky et al., 2023 (v1)

AI-assisted references may not be independent



A reference described as human input may depend on AI outputs.

Detecting AI use is not assessing quality

Veselovsky et al.: text classification and keystroke analysis estimate assistance.

Pangram: reports high-accuracy detection from text alone.

Was AI used? Does the reference support the intended evaluation? These are different questions.

Veselovsky et al., 2023 (v1) Pangram

Discussion 1: What makes this reference trustworthy?

Hypothetical

A lab needs independent human reference summaries, but workers may use LLMs. Choose one intervention: detect AI use, change pay/workload, or redesign the task. What evidence would convince you that it improves the reference data?

Related Students Questions: Will and Paulina

Discuss with classmates; record your own response to Question 1.



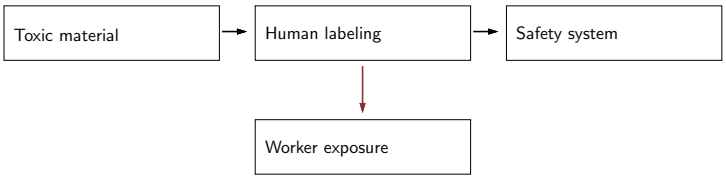
Roadmap

What can this data establish?

Who bears the cost of producing it?

How does AI change future data?

Safer outputs can depend on hazardous work



A model can become safer for users while producing its safety data remains hazardous.

Perrigo, TIME, 2023

Workers reported harm and constrained support

TIME: Kenyan Sama workers labeled toxic text for OpenAI safety systems.

Reported take-home pay: roughly USD 1.32-2 per hour.

Four interviewed workers described psychological harm and counseling constrained by productivity demands.

Sama disputes their accounts of counseling access.

Perrigo, TIME, 2023

Ending a contract does not erase its effects

February 2022

Sama ends its OpenAI work eight months early.

January 2023

Sama announces an exit from all content moderation work by March.

Contract termination is not the same as worker recovery.

Perrigo: OpenAI work Perrigo: moderation exit

Outsourcing changes control and accountability

OpenAI → Sama → Workers

Contract rate: USD 12.50 per hour to Sama; reported take-home of USD 1.32-2.

Possible considerations: costs; legal and regulatory oversight.

Who sets worker protections, and who checks conditions on the ground?

Perrigo, TIME, 2023

Discussion 2: Who can make the work safe?

Hypothetical

A lab proposes LLM labeling with humans reviewing only difficult cases. Would you approve the change? Identify the main worker-safety risk and who should be responsible for addressing it.

Related student questions: Hudson and Yujen

Discuss with classmates; record your own response to Question 2.



Roadmap

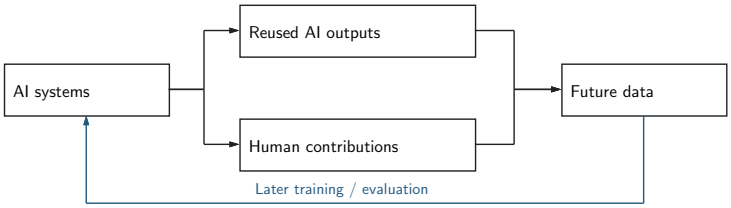
What can this data establish?

Who bears the cost of producing it?

How does AI change future data?

AI can change its own future data supply

Data endogeneity: AI influences the production of future training or evaluation data.



Private AI use can impose public costs

Externalities: costs or benefits that fall on others without being fully accounted for by decision-makers.

Ask a chatbot → Receive a private answer

Possible effect: fewer public contributions for other people to use.

Public contributions fell after ChatGPT's release

Study window: the six months following ChatGPT's release.

Comparisons across programming and mathematics forums.

Activity declined; contributions fell across experience levels.

del Rio-Chanona et al., 2024

Access differences provide comparison settings

The study uses differences in access, not only a before/after trend.

Greater access

Stack Overflow

More limited access

Russian- and Chinese-language counterparts

Compare whether declines are larger where ChatGPT was more available.

del Rio-Chanona et al., 2024

Language differences provide a second test

A separate analysis compares programming languages.

More popular language → Richer training data → Expected stronger answers

Do contribution changes differ where ChatGPT is expected to be more capable?

del Rio-Chanona et al., 2024

Stable peer ratings do not establish community health

Contribution activity

Declined across experience levels.

Peer-voted post quality

No significant change detected.

If the people voting change, does a vote mean the same thing before/after ChatGPT?

del Rio-Chanona et al., 2024

Expert contributions can also be unpaid

Academic collaborators help explain results from frontier AI models.

Scientific explanation → Access, visibility, recognition

OpenAI describes unpaid collaborations as supporting independence.

OpenAI: academic communities

Recognized status shapes rights and protections

Employee, contractor, volunteer: the recognized role matters.

Rights and protections depend on where people live and how their work is classified.

Institutions influence both the production of data and its working conditions.

Discussion 3: A healthy knowledge commons?

> Suppose contributions fall after ChatGPT, but vote scores remain stable. What evidence would convince you that the public knowledge resource is getting worse?

Discuss with classmates; record your own response to Question 3.

> Related student questions: Jolie, Orhun, and Justin

del Rio-Chanona et al., 2024

Provenance links evidence, costs, and future data

Useful data need not imply safe work.

Safer outputs need not imply safe production.

Private benefits need not capture public effects.

> Ask what the data supports, who bears its costs, and how its production is changing.